

Rahul Vimalkanth

rahulvimalkanth@gmail.com

GitHub: xoxo121

LinkedIn: rahul-vimalkanth

EDUCATION

Indian Institute of Technology, Madras | *B.Tech in Electrical Engineering*
Chennai Public School | (CBSE)

CGPA: 7.96 | 2027
Grade: 95% | 2023

PUBLICATIONS

- [1] *Coupling-Aware Planner Exploration for Shared-Workspace Multi-Manipulator Motion Planning*. Accepted at the Xplore Workshop @ IEEE International Conference on Robotics and Automation (ICRA) 2026.
- [2] *Enhancing Financial RAG with Agentic AI and Multi-HyDE: A Novel Approach to Knowledge Retrieval and Hallucination Reduction*. Accepted at FinNLP @ Empirical Methods in NLP (EMNLP) 2025.
- [3] *Lensless Gaze Is Not Private by Default: Auditing Identity Leakage Across Disclosure Surfaces*. Currently Under Review, PFATCV Workshop @ European Conference on Computer Vision (ECCV) 2026.
- [4] *Gaussian Splats Meet Motion Planning: Single-Camera Object Tracking for Robot Autonomy*. Submitted to the IEEE International Conference on Robotics and Automation (ICRA) 2026.
- [5] *When LLMs Orchestrate but Do Not Compute: Tool-Grounded Agents over Structured Streaming Financial Data*. Submitted to SURGeLLM Workshop @ Association for Computational Linguistics (ACL) 2026.

INDUSTRIAL & RESEARCH EXPERIENCE

Software Engineering Intern, Adobe

May '26 – Jul '26

Responsive Layout Recommendation Engine for Adobe Captivate

- Researched **responsive layout generation** via **document understanding** and **mathematical constraint optimization**, translating theoretical research insights into a highly scalable, production-ready system architecture.
- Architected **Prism**, a **semantic layout engine** combining graph-based document reasoning with deterministic geometric planning and bounded AI refinement to precisely generate **responsive mobile interfaces**.
- Implemented the inference engine across **C++**, **TypeScript**, **React**, and browser rendering systems, resolving complex cross-stack integration challenges while building a highly reusable **unified framework**.

Frontier AI Fellow, Pathway

Jul '26 – Present

Memory-Centric AI Systems and Long-Horizon Intelligence

- Selected as an inaugural Frontier AI Fellow to systematically investigate **next-generation memory architectures** for **long-context reasoning**, **long-horizon decision making**, and dynamically adaptive AI systems.
- Developed a unified theoretical framework for **memory update mechanisms**, connecting **fast-weight programming**, **RAG**, **linear attention**, **recurrent memory**, and **test-time learning** via a state-update formulation.
- Studied **persistent memory architectures** for **agentic and embodied AI**, rigorously emphasizing knowledge consolidation, dynamic belief revision, and scalable reasoning over continuously evolving environments.

Research Collaborator, IRT @ AIISC, University of South Carolina

May '26 – Jul '26

Neurosymbolic Reinforcement Learning with Editable Specifications | Advisor: Prof. Amit Sheth

- Developed **SpecEdit**, a **neurosymbolic RL framework** conditioning embodied policies on editable graphs of rules, constraints, and preferences to fully enable **zero-shot task adaptation** without policy retraining.
- Designed **conflict-aware dynamic shielding** to actively detect infeasible specification combinations, attribute blocked actions to specific constraints, and perform minimal, **auditable specification repairs**.
- Built **belief-grounded execution** for **ALFWorld agents** by maintaining persistent symbolic world states and grounding runtime constraints against physical object properties, action histories, and partial observations.

Research Intern, Robotics Research Center, IIIT Hyderabad

Feb '26 – May '26

Sensor-Robust Visuomotor Control for Embodied Navigation | Advisor: Prof. Sourav Garg

- Investigated **sensor-configuration robustness** in **visuomotor navigation policies**, systematically analyzing how camera height, pitch, and intrinsic perturbations affect **waypoint prediction and closed-loop control**.
- Built a controlled data-generation and evaluation pipeline for sensor-perturbed observations, strictly preserving the underlying semantic maps, spatial trajectories, **planning targets**, and precise **path navigation costs**.
- Evaluated invariance-focused training strategies for substantially improving **cross-configuration generalization**, rigorously benchmarking performance via **navigation success**, **shortest path length**, and **collision metrics**.

Young Research Fellow, IIT Madras

Aug '25 – Aug '26

Structured Knowledge Distillation for Prototype-Based Industrial Anomaly Localization

Collaboration with KLA Corporation | Advisors: Prof. Kaushik Mitra & Dr. Pradeep Ramachandran

- Developed **ProtoRank-KD**, a highly streamlined **knowledge-distillation framework** engineered specifically for executing efficient **prototype-based industrial anomaly localization** within production environments.
- Proposed **transport-guided representation transfer** to strictly preserve localization-relevant normality structures during the simultaneous compression of deep functional backbones and memory-based feature prototypes.

- Retained **99.2% of teacher P-AUPRO** accuracy while concurrently achieving $\sim 4\times$ **overall model compression** alongside a highly measurable **2.29 $\times$ acceleration in inference latency**.

Undergraduate Researcher, Computational Imaging Lab, IIT Madras Nov '24 – Jul '26

Lensless Gaze Estimation and Privacy-Risk Auditing | Advisor: Prof. Kaushik Mitra

- Developed a principled analytical framework for rigorously auditing **identity leakage** in **lensless gaze sensing** across optical measurement, digital representation, persistent storage, and output disclosure surfaces.
- Demonstrated that visually unintelligible lensless measurements remain **highly identifying**, empirically challenging the prevailing industry assumption that physical optical hardware obfuscation inherently **guarantees privacy**.
- Proposed **GazeSplit**, a secure public-private feature representation minimizing identity leakage via **semantic tokenization** while preserving crucial downstream utility for **highly accurate gaze estimation tasks**.

Robotics Intern, Robert Bosch Center for Cyber-Physical Systems, IISc Bangalore May '25 – Sep '25

Pose Estimation using Dynamic 3D Gaussian Splats | Advisor: Prof. Pradipta Biswas

- Engineered a sophisticated system for **real-time pose estimation** of dynamic objects using **3D Gaussian Splats** streamed from infrastructure cameras, enabling highly robust **autonomous robot motion planning**.
- Guided foundational research directions through comprehensive literature analyses encompassing **panoptic segmentation, semantic 3D Gaussian Splatting, precise pose estimation, and planar reconstruction**.
- Authored a comprehensive technical paper detailing a **novel scene retrieval and six-degree-of-freedom pose estimation method** leveraging **explicit neural radiance fields** for analyzing dynamic environment states.

AI Intern, Tattvam AI May '24 – Jul '24

- Investigated **ARC-AGI abstract reasoning** and **program-synthesis benchmarks** explicitly designed to meticulously expose algorithmic generalization failures far beyond standard language-model pattern matching capabilities.
- Developed and rigorously evaluated reasoning-oriented solution strategies while significantly improving model adaptation efficiency through the systematic application of advanced **parameter-efficient fine-tuning techniques**.

POSITIONS OF RESPONSIBILITY

Problem Statement Lead, Inter-IIT TechMeet 14.0 Oct '25 – Dec '25

- Spearheaded a **team of 8** to architect a **real-time agentic treasury orchestration system** using **Pathway**, seamlessly combining **automated cash flow forecasting** with dynamic multi-asset investment execution.
- Engineered a **multi-agent framework** via **LangGraph** and **Pathway** for asynchronous fusion of high-frequency market ticks and macro data, integrating **online streaming ML** for **dynamic interest rate forecasting**.
- Developed specialized **Bond Pricing** and **Investment agents** proactively generating actionable buy/sell signals accompanied by **fully explainable rationales** under shifting monetary policies and market volatility.

Contingent Member, Inter-IIT TechMeet 13.0 Oct '24 – Dec '24

- Contributed to a dynamic **agentic RAG framework** leveraging **multi-HyDE** and deterministic state machines to ensure high-accuracy, rigorously **low-hallucination document retrieval** across complex SEC filings.
- Built custom caching and **95% accurate intent differentiation** for financial RAG pipelines, significantly improving numerical precision, temporal alignment, and high-volume semantic retrieval using **FAISS** and **LRU caches**.
- Developed a highly robust **user variable storage architecture** specifically engineered to reliably manage **stateful updates** to user preferences and dynamically retrieved data across multiple sequential interaction sessions.

Project Member, WebOps & Blockchain Club, CFI May '24 – Mar '25

- Architected a **multi-agent autonomous AI code assistant** utilizing **LangGraph**, enabling the system to independently write, execute, iteratively debug, and **fully validate complex source code** from natural language prompts.
- Engineered comprehensive system capabilities encompassing secure **local file system interaction**, automated syntax and complex logic **error correction**, and seamless integration for **live website deployment**.

SKILLS & COURSEWORK

Programming:	Python, C/C++ , Verilog, ARM/AVR Assembly
Frameworks & ML:	PyTorch, TensorFlow, CUDA, OpenCV, LangChain, LangGraph, Hugging Face, Scikit-Learn
Tools & Cloud:	Docker, Git/GitHub, Weights & Biases (WandB), Bash, Unslloth
Courses:	Deep Learning, Reinforcement Learning, Natural Language Processing, Motion Planning, Automated Planning & Learning, Federated Learning & Distributed ML, Computational Photography, Control Systems, Computer Organization, Probability, Signals & Systems
Certifications:	ML & DL Specialization (Andrew Ng), OpenCV Bootcamp

SCHOLASTIC ACHIEVEMENTS

- Secured **AIR 1728** in JEE Advanced 2023 (**top 1%** among 200,000 highly competitive candidates).
- Ranked in the **top 0.5 percentile** of JEE Main 2023 among a nationwide applicant pool exceeding 1.4 million.
- Secured **AIR 372** in KVPY 2022, earning direct undergraduate admission offers from premier research institutions.
- Achieved **State Rank 2** in the NTSE Stage-I 2021 and subsequently placed in the **top 800 nationally**.
- Awarded a prestigious **Certificate of Merit** in the Indian Olympiad Qualifier for Mathematics (IOQM).